

MAE 280B - Linear Control Design
Assignment #1

Problem 1

(Linear Optimal Control by Jeffrey B. Burl, pp. 227)

Given the system:

$$\dot{x}(t) = 2x(t) + u(t)$$

and the cost function:

$$J(x(t), y(t)) = \int_0^{\infty} 2x^2(t) + u^2(t) dt$$

- a) Compute the Riccati equation for this control problem.
- b) Solve for the optimal feedback gain.
- c) What is the pole of the steady-state, closed-loop system?
- d) How does this pole change if the state weighting is increased by a factor of 10?
- e) If $x(0) = 100$, what is the optimal cost?

Problem 2

(Optimal Control and Estimation by Robert F. Stengel, pp.616)

A second-order robot arm is controlled by a first-order actuator, resulting in the following mechanical model:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -\omega_n^2 & -2\zeta\omega_n & \omega_n^2 \\ 0 & 0 & -1/\tau \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1/\tau \end{bmatrix} u(t)$$

Compute the gain matrix, $\mathbf{K}$, using the algebraic Riccati equation, given

$$Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad R = 1 \quad \zeta = 0.3 \quad \omega_n = \pi \quad \tau = 1$$

Problem 3

(Optimal Control and Estimation by Robert F. Stengel, pp.617)

A fourth-order model of an aircraft's longitudinal motions can use axial and normal velocities, pitch rate, and pitch angle as state vector components and elevator and throttle as control vector components. The corresponding system matrices for a high-performance aircraft flying at an altitude of 20,000 ft and airspeed of 700 ft/s are as follows:

$$A = \begin{bmatrix} -0.04 & 0.1 & 54.3 & -32.1 \\ -0.02 & -0.87 & 698 & -2.62 \\ 0 & 0.003 & -0.74 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$
$$B = \begin{bmatrix} 0 & 14.5 \\ 0 & 0 \\ -7.22 & 0 \\ 0 & 0 \end{bmatrix}$$

Examine the effects of cost function weighting on the initial-condition response and eigenvalues of the aircraft model. The nominal values of $\mathbf{Q}$ and $\mathbf{R}$ are as follows:

$$\mathbf{Q} = \text{Diag}[1, 1, 1, 1]$$

$$\mathbf{R} = \text{Diag}[100, 1]$$

- a) Individually increase and decrease the elements of $\mathbf{Q}$ by factors of 10. Compute the effects of these modifications on the closed-loop initial-condition response. For variations in q_{ii} , choose the initial condition as $x_i(0) = 1$, $x_j = 0$ ($j \neq i$).
- b) Individually and jointly increase and decrease the elements of $\mathbf{R}$ by factors of 10. Compute and plot the closed-loop eigenvalues.